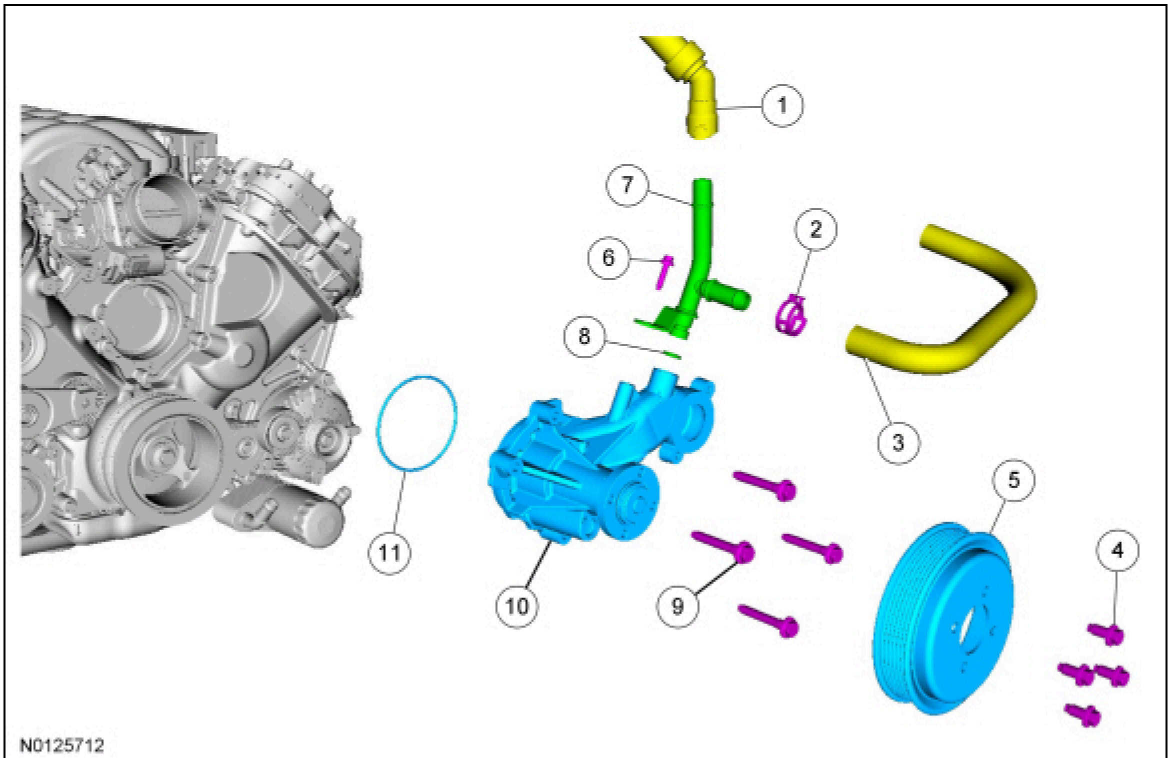
Coolant Pump — 5.0L

Ford Ecat Electronic Parts Catalog

Material

Item	Specification
Motorcraft® Metal Surface Prep ZC-31-B	—
Motorcraft® Orange Antifreeze/Coolant Concentrated VC-3-B (US); CVC-3-B2 (Canada)	WSS-M97B44-D



Item	Part Number	Description
1	1847218472	Heater outlet hose
2	—	Degas bottle-to-engine hose clamp (part of 8C350)
3	8C3508C350	Degas bottle-to-engine hose
4	N806282N806282	Coolant pump pulley bolt (4 required)
5	85098509	Coolant pump pulley
6	W503277W503277	Heater outlet tube bolt - 10 Nm (89 lb-in)
7	1866318663	Heater outlet tube
8	85658565	Heater outlet tube O-ring seal
9	W714925W714925	Coolant pump bolt (4 required)
10	85018501	Coolant pump

11	85078507	Coolant pump O-ring seal (part of 8501)

Removal

All vehicles

1. Drain the cooling system. For additional information, refer to Cooling System Draining, Filling and Bleeding in this section.
2. Remove the air cleaner outlet pipe. For additional information, refer to Section 303-12.
3. Loosen the 4 coolant pump pulley bolts.
4. Remove the accessory drive belt. For additional information, refer to Section 303-05.

Vehicles with an A/C compressor belt tensionser

5. **NOTICE: Remove the A/C compressor belt to prevent coolant contamination of the belt.**

   Remove the A/C compressor belt.

Vehicles without an A/C compressor belt tensioner

6. **NOTICE: Cover the A/C compressor belt to prevent coolant contamination of the belt.**

   Completely cover the A/C compressor belt with waterproof plastic.

All vehicles

7. Remove the thermostat housing. For additional information, refer to Thermostat Housing — 5.0L in this section.
8. Remove the bolts and the coolant pump pulley.
9. Disconnect the heater outlet hose from the heater outlet tube.
10. Release the clamp and disconnect the degas bottle-to-engine hose from the heater outlet tube.
11. Remove the 4 bolts and the coolant pump.
12. If a new coolant pump is being installed, remove the bolt and the heater outlet tube.
    - Remove and discard the O-ring seal.

Installation

All vehicles

1. If a new coolant pump is being installed, install a new O-ring seal on the heater outlet tube and lubricate it with clean engine coolant.
2. Install the heater outlet tube and the bolt.
   - Tighten to 10 Nm (89 lb-in).
3. Inspect the sealing surfaces and clean with metal surface prep. Follow the directions on the packaging.
4. **NOTICE: Align the bolt holes with the bosses prior to insertion of the coolant pump and insert the pump straight into the coolant pump cavity. Do not rotate the coolant pump once installed in the coolant pump cavity or the O-ring seal can be damaged, causing the coolant pump to leak.**

   NOTE: *Install a new O-ring seal and lubricate it with clean engine coolant.*

Install the coolant pump and the bolts
- Tighten the coolant pump bolts in 2 stages.
  - Stage 1: Tighten to 20 Nm (177 lb-in).
  - Stage 2: Tighten an additional 60 degrees.

5. Connect the degas bottle-to-engine hose to the heater outlet tube and position the clamp.

6. Connect the heater outlet hose to the heater outlet tube.

7. Position the coolant pump pulley and install the bolts finger tight.

8. Install the thermostat housing. For additional information, refer to Thermostat Housing — 5.0L in this section.

Vehicles without an A/C compressor belt tensioner

9. Remove the plastic from the A/C compressor belt.

Vehicles with an A/C compressor belt tensioner

10. Install the A/C compressor belt. For additional information, refer to Section 303-05.
    - Rinse off any coolant from the accessory drive pulleys using clear water prior to belt installation.

All vehicles

11. Install the accessory drive belt. For additional information, refer to Section 303-05.
    - Rinse off any coolant from the accessory drive pulleys using clear water prior to belt installation.

12. Tighten the coolant pump pulley bolts in a criss-cross pattern.
    - Tighten to 25 Nm (18 lb-ft).

13. Install the air cleaner outlet tube. For additional information, refer to Section 303-12.

14. Fill and bleed the cooling system. For additional information, refer to Cooling System Draining, Filling and Bleeding in this section.